

physics modelling agent

first prototype

Janis Jacobi
Niels Schröter

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outline

- 1 goal of the project & aim of this prototype
- 2 architecture
 - constraints
 - crewai
- 3 example workflow
 - input
 - the Edelstein effect
 - pipeline
 - agent & task configuration
- 4 results
- 5 summary

the general goal

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The model takes a research request as input and suggest a model that answers which answers the research question

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Gather information related to the question

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Suggest a simple model

```
graph TD; A[Gather information related to the question] --> B[Suggest a simple model]
```

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Create a numerical implementation

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Suggest a simple model

Create a numerical implementation

Summarise the results and output them for a human

this prototype

what matches the final goal

- information is extracted
- real sources are used
- a model is suggested
- a numerical implementation is suggested
- the output seems reasonable

this prototype

what matches the final goal

- information is extracted
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what is still missing

- the concrete sources have to be provided manually
- the numerical implementation is not a direct python file, just the code
- the results are not summarised in a easy to read shape

constraints

architectural foundations

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- only use open source software
- only use free software (no payed subscriptions)

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design choices

- use SAIA from GWDG[7]
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 - do not re-implement the entire logic for how to orchestrate different AI's
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Figure: crewai logo [3]

crewai: core components

Agents[1]

accesses a LLM to work on a given topic

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- accesses the LLM via specific object (allows to use SAIA or any other provider, specifies which model to use)
- behaviour can be defined via parameters (name, goal, backstory, temperature)
- tools can be added to allow advanced abilities (web search, code execution)

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describe a concrete action to execute

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Tasks[4]

describe a concrete action to execute

- has task description and expected output parameters
- gets an agent object assigned (execution merges agent and task description)

crewai: orchestration

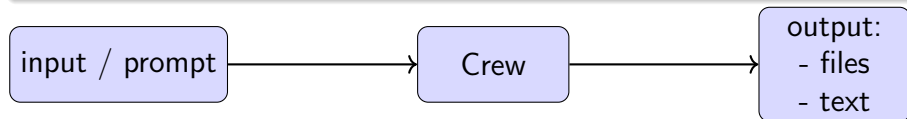
Crew[2]

this puts it all together and orchestrates order of task executions

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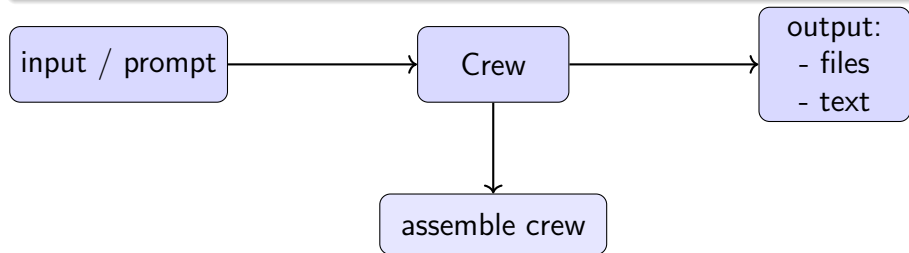
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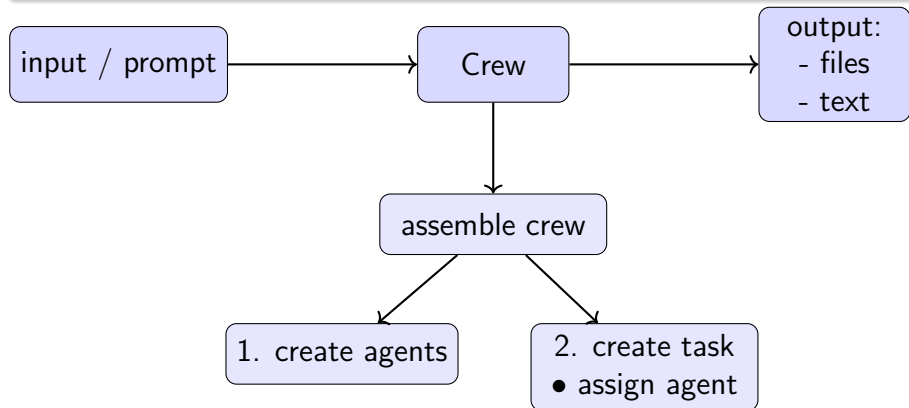
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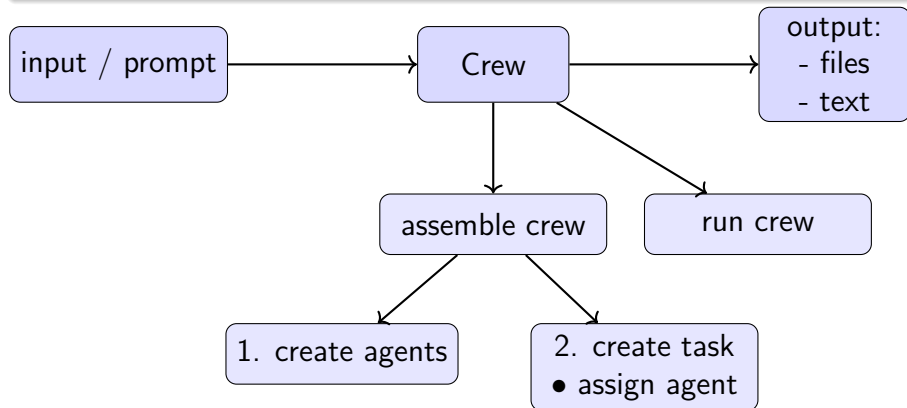
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crews internal logic

upon creation of “crew” object:

- creates agent and task objects from config-files

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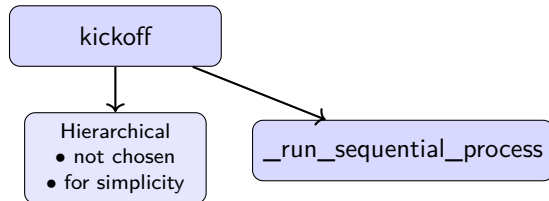
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kickoff

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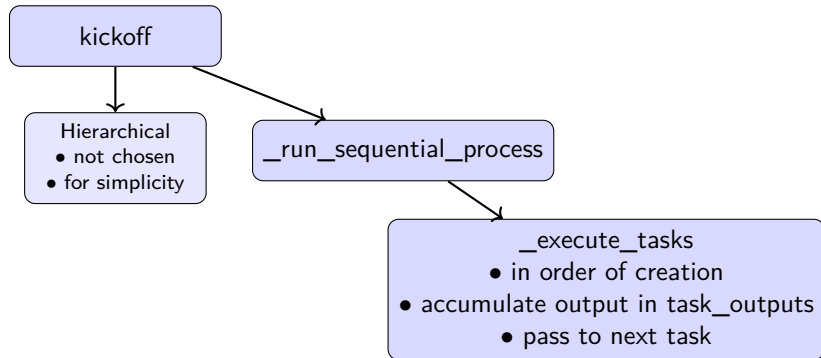
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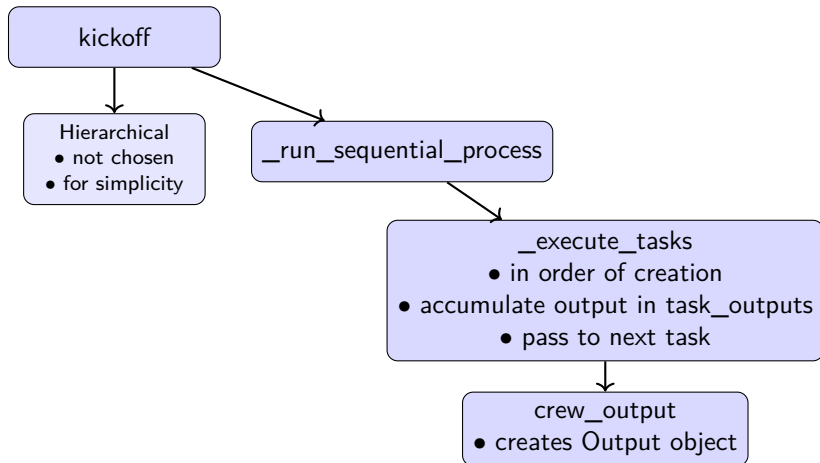
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example: input/modelling request

input

“Calculate the Edelstein effect for a Kramers-Weyl fermion at the Gamma point of the Brillouin zone. Compute the magnetization magnitude and direction of different directions and magnitudes of the applied electric field.

Consider how the result depends on relevant parameters of the model (e.g. chirality, fermi velocity)”

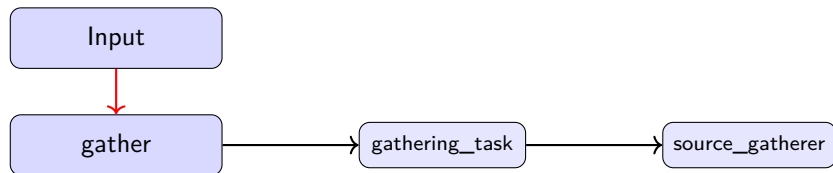
topic: Edelstein Effect

year: 2026

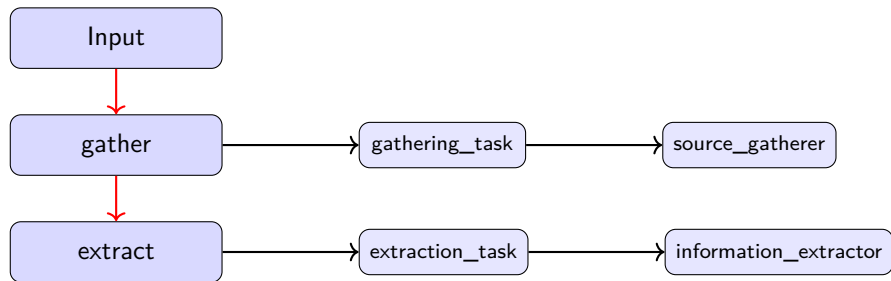
pipeline

Input

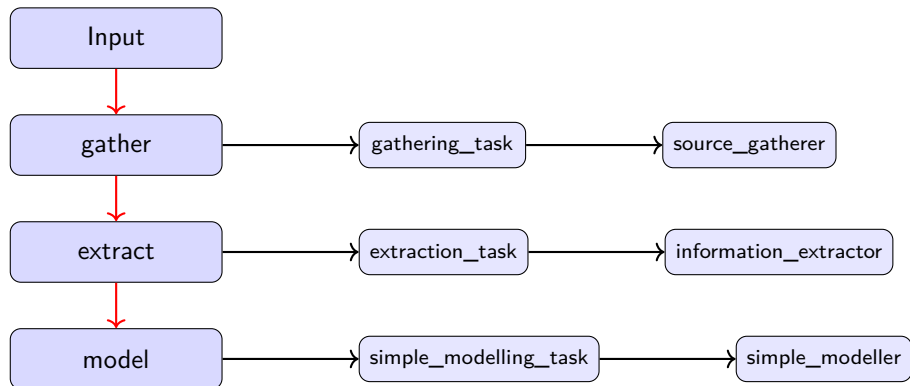
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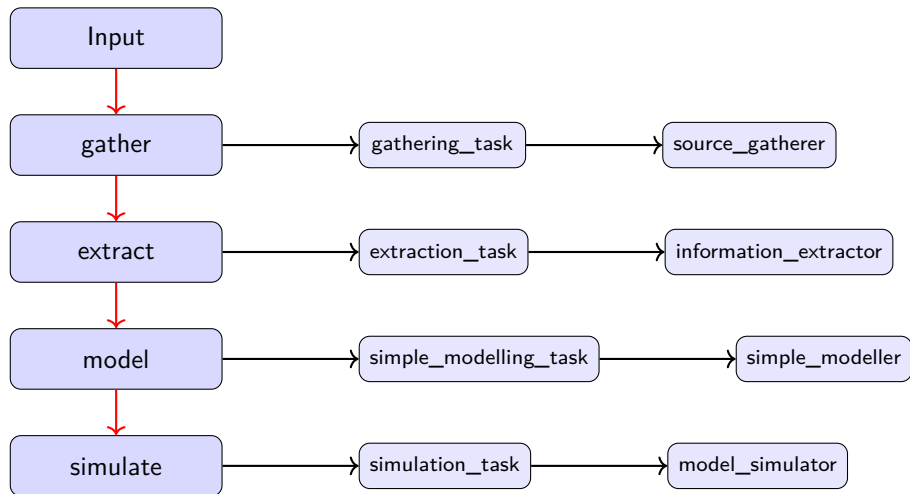
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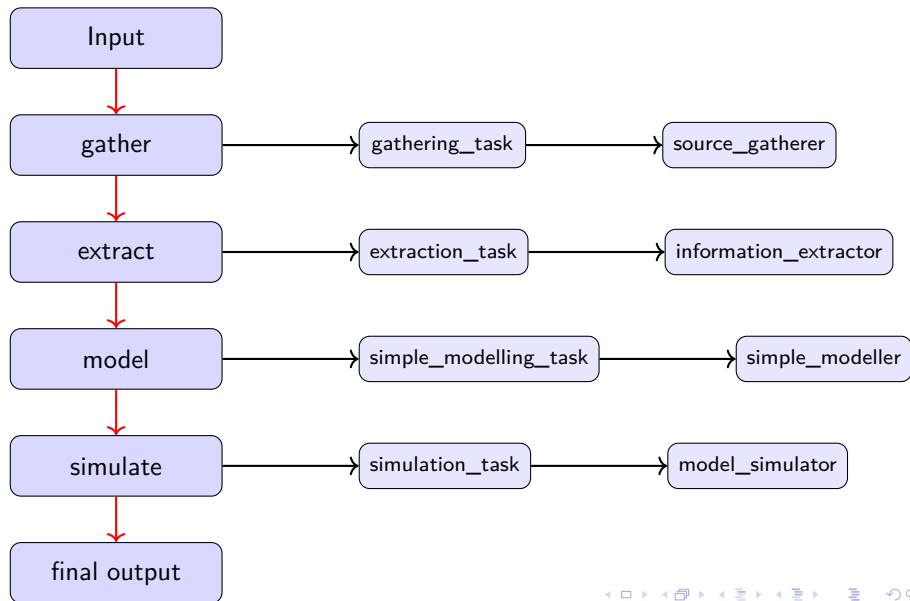
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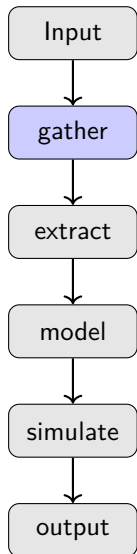
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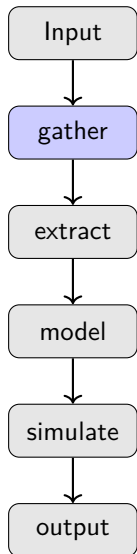
agent: find relevant scientific papers



gather

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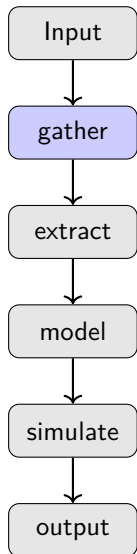
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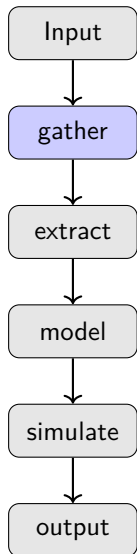
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- tools = ScrapeWebsiteTool
 - urls to scientific papers; example : “Spin and orbital Edelstein effects in a two-dimensional electron gas: Theory and application to SrTiO3 interfaces”[10]



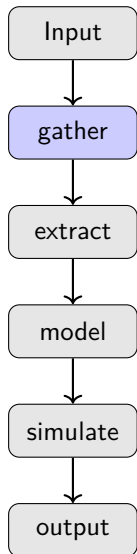
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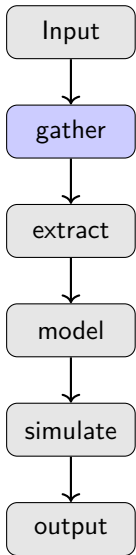
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- output_file: create a markdown file





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3. Spin and orbital Edelstein effects in a two-dimensional electron gas: theory and application to SrTiO₃ interfaces

Authors: A. Johansson, B. Göbel, J. Henk, M. Bibes, I. Mertig

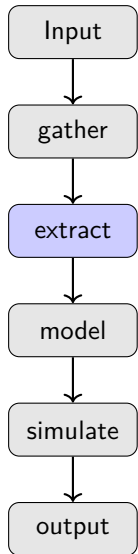
Where it was found: Physical Review Research, Volume 3, 013275 (2021)

Summary: This paper provides theoretical treatment of both spin and orbital Edelstein effects in two-dimensional electron gases, with specific application to SrTiO₃ interfaces. It develops the microscopic theory connecting Rashba-type spin-orbit coupling to current-induced spin polarization and orbital angular momentum effects.

Figure: source gatherer example output

extract

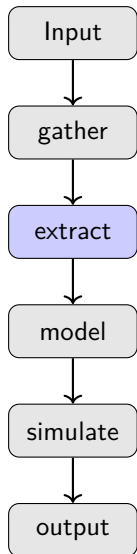
agent: extract information using the sources of gather



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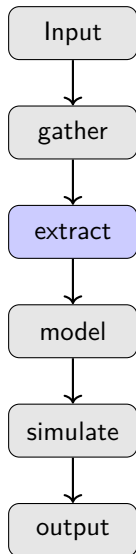


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task: test and mathematical formulas; create md

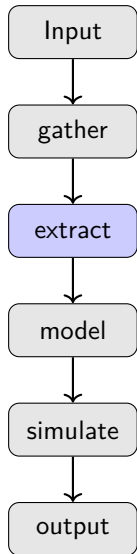


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4.1 Energy Dispersion

For a KW fermion at Gamma point: $\epsilon_{k, \pm} = \pm \chi v_F |k|$

The two bands correspond to opposite chirality states.

4.2 Spin Texture

The spin expectation value for states with momentum k : $\langle S(k) \rangle = \frac{\hbar}{2} \frac{k}{|k|}$

This is the characteristic radial spin texture of Kramers-Weyl fermions.

4.3 Current-Induced Spin Polarization

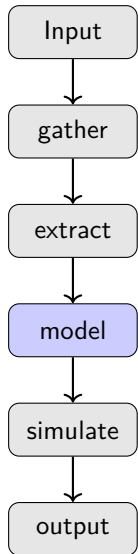
Under applied electric field E , the shift in distribution function creates net spin polarization:

$$S = \int \frac{d^3k}{(2\pi)^3} \langle S(k) \rangle \left(-\frac{\partial f_0}{\partial \epsilon} \right) e \tau v_k \cdot E$$

Figure: information extractor example output

model

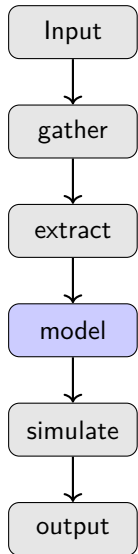
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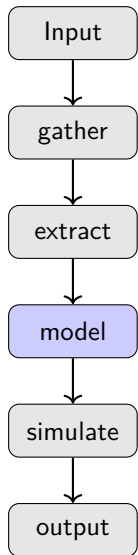
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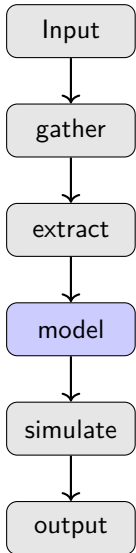
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task: mathematical description with textual explanation

3. Spin Texture

The spin expectation value for momentum $\langle \mathbf{S} \rangle$ is: $\langle \mathbf{S} \rangle = \frac{\hbar}{2} \chi \frac{\langle \mathbf{k} \rangle}{|\langle \mathbf{k} \rangle|}$

4. Edelstein Effect Calculation

Using the Boltzmann transport approach, the spin polarization $\langle \mathbf{S} \rangle$ is proportional to the electric field $\langle \mathbf{E} \rangle$: $\langle \mathbf{S} \rangle = \lambda \langle \mathbf{E} \rangle$ where λ is the Edelstein coefficient.

5. Edelstein Coefficient

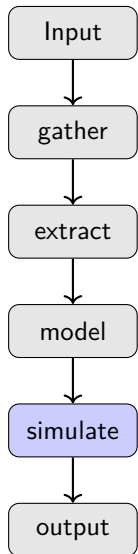
The coefficient is given by: $\lambda = \frac{e^2 \tau \chi v_F k_F^2}{3 \pi^2 \hbar^2}$ where:

- e is the elementary charge.
- τ is the relaxation time.
- k_F is the Fermi wavevector.

Figure: model maker example output

simulation I

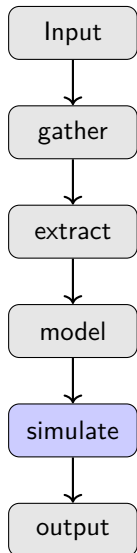
agent: create a numerical implementation of the model



simulation I

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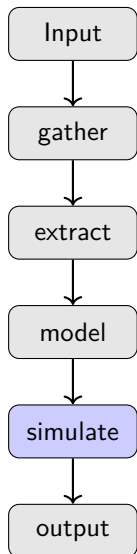
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simulation I

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task: create executable python code

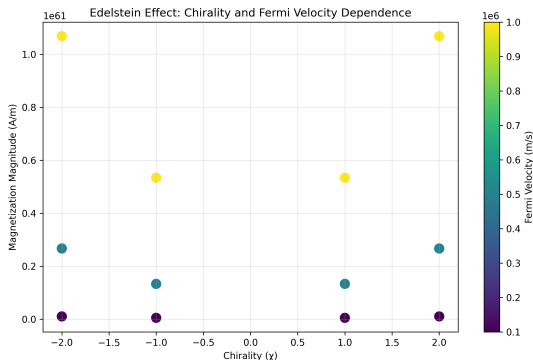
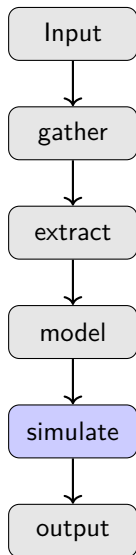
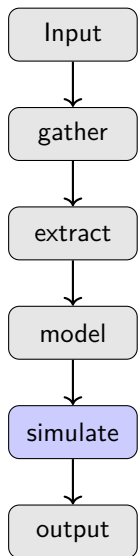


Figure: Magnetisation dependence; simulation plot

simulation II

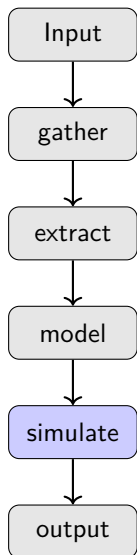
result: **executable code**



simulation II

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- calculate magnetisation from formula $M = -\frac{g \cdot \mu_B}{\hbar} * \vec{S}$



simulation II

result: **executable code**

- calculate magnetisation from formula $M = -\frac{g \cdot \mu_B}{\hbar} * \vec{S}$
- no numerical methods or step by step calculation

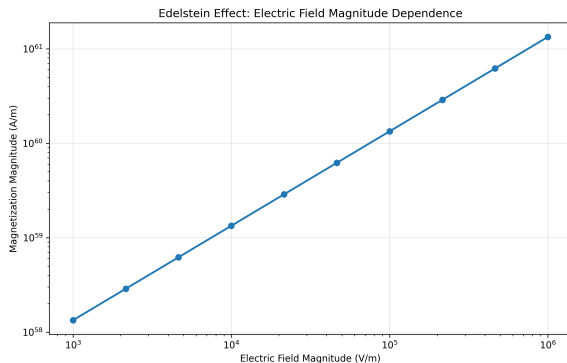
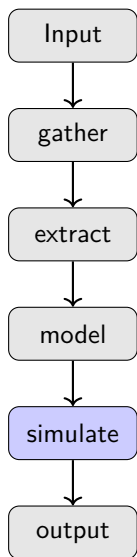


Figure: Electric field dependence; simulation plot

the key insights

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- existing & up to date sources

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pipeline improvements

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pipeline improvements

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 - this could lead to the crew actually making a **simple** model first
 - as of right now: it seems as default agent behaviour is overwritten → crew tries to solve the given question
- get an real numerical implementation (using time steps and calculation from previous ones)
- better output format: convert markdown to latex

parameter configuration

temperature

test how changes affect output

parameter configuration

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test how changes affect output

models

find better benchmarks for which models to use (for
“devstral-2-123b-instruct-2512” no scientific bench marking found)

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tools

- use a real web search tool or a RAG DB
- equip the simulator with a tool for python execution

parameter configuration

temperature

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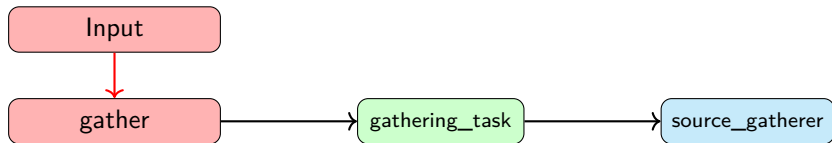
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benchmark the crew

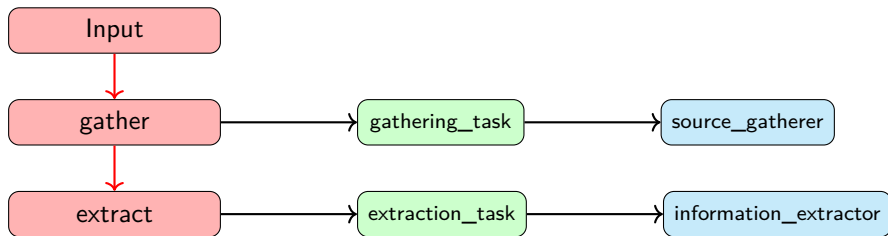
use specific problem sets[5] from critpt[12]

Input

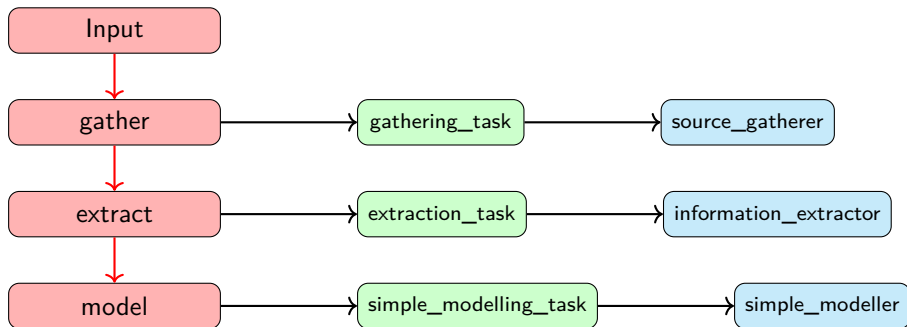
summary/status quo

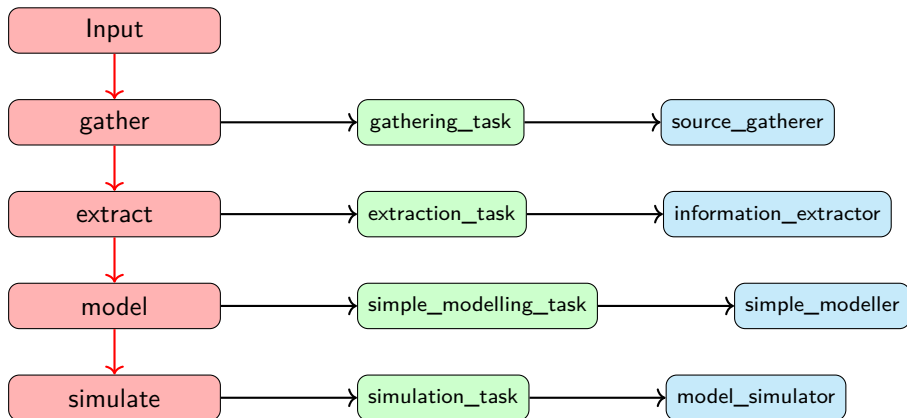


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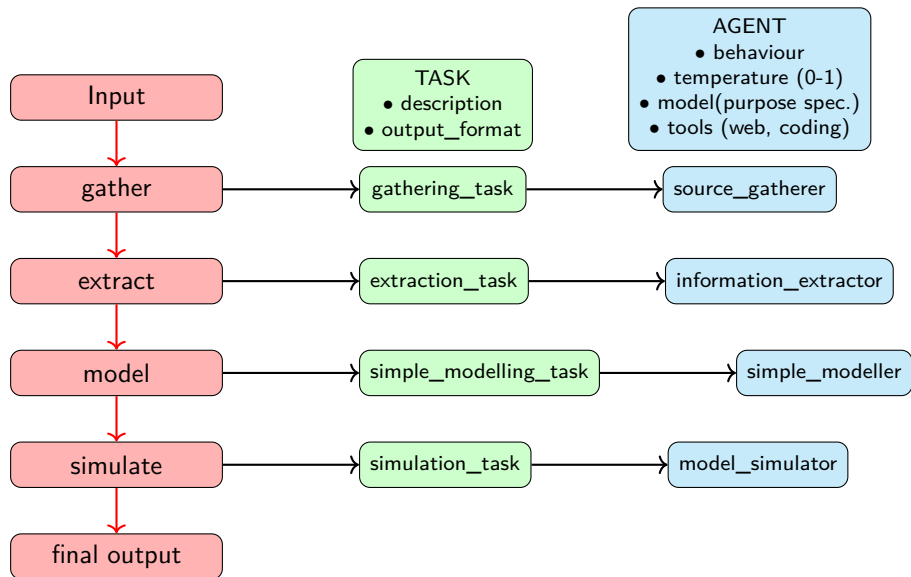


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- [1] crewai. *core.py*. 04/22/2026. URL: <https://github.com/crewAIInc/crewAI/blob/main/lib/crewai/src/crewai/agent/core.py> (visited on 04/22/2026).
- [2] crewai. *crew.py*. 04/22/2026. URL: <https://github.com/crewAIInc/crewAI/blob/main/lib/crewai/src/crewai/crew.py> (visited on 04/22/2026).
- [3] crewai. *crewAi*. 04/22/2026. URL: <https://crewai.com/> (visited on 04/22/2026).
- [4] crewai. *task.py*. 04/22/2026. URL: <https://github.com/crewAIInc/crewAI/blob/main/lib/crewai/src/crewai/task.py> (visited on 04/22/2026).

- [5] critPt. *challenges*. 04/22/2026. URL:
https://github.com/CritPt-Benchmark/CritPt/tree/main/data/public_test_challenges
(visited on 04/22/2026).
- [6] Gesellschaft für wissenschaftliche Datenverarbeitung mbH Göttingen. *AcademicCloud*. 04/21/2026. URL:
<https://id.academiccloud.de> (visited on 04/22/2026).
- [7] Gesellschaft für wissenschaftliche Datenverarbeitung mbH Göttingen. *SAIA*. 04/21/2026. URL:
<https://docs.hpc.gwdg.de/services/ai-services/saia/index.html> (visited on 04/22/2026).

- [8] Hugging Face. *Devstral 2 123B Instruct 2512*. 04/22/2026. URL: <https://huggingface.co/mistralai/Devstral-2-123B-Instruct-2512> (visited on 04/22/2026).
- [9] Daya Guo et al. “DeepSeek-R1 incentivizes reasoning in LLMs through reinforcement learning.” In: *Nature* 645.8081 (2025), pp. 633–638. ISSN: 1476-4687. DOI: [10.1038/s41586-025-09422-z](https://doi.org/10.1038/s41586-025-09422-z). URL: <http://dx.doi.org/10.1038/s41586-025-09422-z>.
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- [11] Qwen Team. *Qwen3.5-Omni Technical Report*. 2026. arXiv: 2604.15804 [cs.CL]. URL: <https://arxiv.org/abs/2604.15804>.
- [12] Minhui Zhu et al. *Probing the Critical Point (CritPt) of AI Reasoning: a Frontier Physics Research Benchmark*. 2025. arXiv: 2509.26574 [cs.AI]. URL: <https://arxiv.org/abs/2509.26574>.

AI usage

most parts of the drawn figures are created in cooperation with [microsoft copilot](#). I take full responsibility for the figures contents.